

HAWK HILLS AGCM, AB, CanadaWMO: **715070**Lat: **57.265N**Lon: **117.298W**Elev: **455**StdP: **95.98**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -35.1	(c) -32.0	(d) -38.8	(e) 0.1	(f) -35.0	(g) -35.5	(h) 0.1	(i) -31.9	(j) 10.4	(k) -3.2	(l) 8.9	(m) -8.8	(n) 1.5	(o) 210	(p) 0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	28.4	16.9	26.6	16.2	24.9	15.5	18.5	25.3	17.6	24.0	16.7	22.8	3.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	12.0	20.7	15.1	11.3	20.2	14.1	10.6	19.1	54.3	25.8	51.2	24.2	48.4	22.9	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	7.6	6.6	-39.0	31.6	2.7	1.5	-40.9	32.7	-42.5	33.6	-44.0	34.4	-45.9	35.5
			-39.1	20.5	2.7	1.3	-41.0	21.4	-42.6	22.2	-44.1	22.9	-46.0	23.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-14.7	-14.4	-8.1	2.3	10.3	14.8	16.7	15.3	10.1	2.6	-9.0	-15.6
	DBStd	13.60	9.13	8.25	8.33	5.66	4.61	3.27	2.62	3.22	4.46	4.91	7.34	8.40
	HDD10.0	3952	766	682	560	236	52	4	1	3	52	234	569	794
	HDD18.3	6376	1024	915	818	481	249	112	63	103	250	488	819	1052
	CDD10.0	645	0	0	0	4	62	148	208	166	53	4	0	0
	CDD18.3	29	0	0	0	0	1	6	13	7	1	0	0	0
	CDH23.3	756	0	0	0	6	70	151	295	184	49	1	0	0
	CDH26.7	154	0	0	0	1	13	31	63	38	9	0	0	0
Wind	WSAvg	2.9	3.1	2.8	2.9	3.3	3.5	2.9	2.5	2.4	2.7	2.8	2.8	2.9
Precipitation	PrecAvg	486	27	18	21	24	46	88	85	57	41	28	27	24
	PrecMax	651	59	31	44	45	90	158	147	107	107	65	57	58
	PrecMin	317	7	6	5	4	15	30	47	19	4	13	11	7
	PrecStd	78	13	7	8	9	19	34	28	21	18	12	12	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.5	6.0	10.6	23.6	28.4	29.9	31.3	30.3	27.9	20.5	9.0	5.7
		MCWB	3.8	2.4	4.9	11.6	14.2	16.9	18.9	18.3	16.5	11.7	5.5	2.1
	2%	DB	4.8	3.7	8.0	17.6	25.2	26.9	28.3	27.4	24.4	15.2	5.6	2.2
		MCWB	2.1	1.1	3.4	8.2	12.5	16.1	17.5	17.0	14.8	8.4	2.9	0.0
	5%	DB	2.6	0.9	5.8	13.8	22.7	24.9	26.5	25.3	21.2	12.7	3.3	-0.7
		MCWB	0.7	-0.6	2.2	6.1	11.2	15.1	16.8	16.3	13.0	7.3	1.1	-2.0
	10%	DB	-0.3	-2.4	3.6	11.4	20.6	22.9	24.7	23.4	18.5	10.1	0.8	-3.7
		MCWB	-1.9	-3.6	0.7	5.3	10.4	14.1	16.2	15.6	11.8	6.0	-0.6	-4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	2.7	5.3	11.7	15.0	18.9	20.4	19.7	17.2	12.4	5.6	2.5
		MCDB	7.1	5.2	10.1	22.7	26.6	25.9	28.2	27.1	26.1	18.5	8.8	5.1
	2%	WB	2.4	1.2	3.8	8.6	13.3	17.3	18.8	18.2	15.3	9.3	3.0	-0.1
		MCDB	4.6	3.2	7.5	16.2	22.6	24.0	25.5	24.5	22.3	14.2	5.4	2.0
	5%	WB	0.8	-0.8	2.3	6.8	12.1	16.2	17.7	17.3	14.0	7.6	1.3	-2.1
		MCDB	2.5	1.2	5.6	12.9	20.4	22.3	24.3	23.4	20.1	11.8	3.1	-0.7
	10%	WB	-1.8	-3.7	0.8	5.4	11.0	15.1	16.9	16.4	12.4	6.2	-0.6	-4.8
		MCDB	-0.2	-2.5	3.4	10.8	19.1	20.9	23.1	21.9	17.6	9.6	0.7	-4.0
Mean Daily Temperature Range	5% DB	MDBR	9.2	11.4	12.8	12.3	15.0	12.9	13.9	13.9	13.7	9.5	8.5	8.8
		MCDBR	11.6	12.2	13.4	16.8	19.9	17.3	17.6	17.5	18.0	14.2	10.7	10.7
		MCWBR	9.5	9.9	9.5	9.2	9.4	8.0	8.2	8.3	9.1	8.5	8.2	9.2
	5% WB	MCDBR	11.5	12.1	13.0	15.8	17.2	14.6	16.2	15.8	17.3	13.5	10.3	10.6
		MCWBR	9.5	10.0	9.1	9.0	8.4	7.2	8.3	8.4	9.3	8.5	8.2	9.2
Clear-Sky Solar Irradiance	taub		0.227	0.254	0.280	0.333	0.347	0.362	0.366	0.351	0.310	0.287	0.240	0.216
	taud		2.283	2.316	2.345	2.362	2.415	2.392	2.403	2.426	2.555	2.569	2.405	2.189
	Ebn at Noon		691	820	888	883	890	877	865	854	851	784	678	581
	Edh at Noon		45	69	89	104	105	109	106	97	73	55	40	37
All-Sky Solar Radiation	RadAvg		0.63	1.57	3.16	4.72	5.62	5.89	5.64	4.61	3.15	1.63	0.68	0.41
	RadStd		0.05	0.13	0.24	0.27	0.44	0.51	0.34	0.38	0.23	0.22	0.06	0.03

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	+1.94	N/A	N/A	N/A	N/A	N/A	+50	N/A

Nomenclature: See separate page